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Development of an IoT Sensor-based System for Soil Monitoring: Conceptual Design and Experiments ⋆

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Abstract:
High-quality forage is essential for efficient livestock production, directly impacting the meat and dairy industries. Improved forage quality enhances yield and nutritional value, ultimately boosting productivity. The DLT-Farming project aims to modernize grass forage production and improve nitrogen use efficiency by integrating data-driven technologies and genomics. This paper presents a proof-of-concept (PoC) soil monitoring system that leverages low-cost IoT sensors to track soil conditions. Hydroponic setup and laboratory experiments evaluate sensor consistency and their ability to detect soil quality indicators under varying nutrient treatments. While the sensors detected some changes, several measurements require careful interpretation due to observed limitations in accuracy. A power consumption analysis revealed that the sensors use less energy than specified in the datasheet. However, their power draw remains constant, regardless of communication with the host computer, highlighting the need for improved power control mechanisms.

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1. INTRODUCTION

Forage crops account for over 70% of the global agricultural area in the 21st century, according to Conant (2010). However, as Capstaff and Miller (2018) points out, research on forage fields lags behind when compared with cereals, fruits, and vegetables. Hoppe and Carlson (2023) reinforces that enhanced forage quality can lead to better market value and increased economic returns for farmers.

In Norway, plant breeding efforts focus on developing varieties adapted to the Nordic climate. However, traditional breeding relies on manual measurements, which are labor-intensive, time-consuming, and costly, driving the need for automation. This has driven advancements in precision agriculture, which aims to enhance efficiency and sustainability through sensing technologies and robotics, as highlighted by King (2017).

The adoption of IoT-based systems in agriculture also has gained significant traction over the last decade due to the increasing affordability of sensors and growing demand for food production. Initial implementations commonly deploy Arduino-based boards, as in Mutyalamma et al. (2020), Aarthi et al. (2023), and Kumar et al. (2024) works. However, Aarthi et al. (2023) also incorporates a Raspberry Pi as a gateway unit, while Rajani et al. (2023) deploys a Raspberry Pi to handle all processing directly. Another widely used approach involves ESP-based boards, such as the ESP32, as used by Postolache et al. (2022), and the ESP8266, as seen in the work of Cheruvu et al. (2023). Some studies, like Ramson et al. (2021), explore the use of fully custom hardware solutions.

LoRaWAN is one of the most widely adopted Wireless Sensor Network (WSN) protocols for agricultural IoT applications due to its long-range and low-power characteristics. Several studies, including those by Ramson et al. (2021), Postolache et al. (2022), and Othaman et al. (2021), leverage LoRaWAN for efficient data transmission. Additionally, Postolache et al. (2022) employs Bluetooth Low Energy (BLE) for short-range communication and system configuration. Zigbee is another relevant WSN protocol, as discussed by Musa et al. (2023) and Ramson et al. (2021), offering a low-power solution for short-range communication. Traditional Wi-Fi networks operating at $2.4GHz$ are also frequently used as a low-cost and readily available entry point for IoT devices.

For data visualization and management, Thingspeak is a commonly used platform, as referenced in the works of Cheruvu et al. (2023), Rajani et al. (2023), and Othaman et al. (2021). Some implementations also incorporate on-board displays and mobile integration for enhanced user interaction. Python remains the predominant language for data handling in most studies.

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A crucial aspect of IoT-based agricultural systems is the precision, reliability, and challenges associated with sensor data. Studies by Morchid et al. (2023), Pascoal et al. (2024), and Musa et al. (2023) address these concerns in their reviews. Many projects rely on single-sensor deployments or limited sampling, restricting insights into data consistency across multiple sensors and over time.

None of the reviewed works investigate the potential for robotic integration, and only a few, such as Ramson et al. (2021), explore the possibility of generating soil maps. However, Schirrmann et al. (2011) presents a use case involving a soil pH mapping tool designed to be towed by a tractor. By integrating robotics, there is an opportunity to enhance scalability and automation, while leveraging IoT sensors to provide insights into soil state and plant health.

As such, the DLT-Farming project aims to enable a data-driven transformation of forage grass farming. By integrating IoT sensors, drone-based multispectral imaging, and mobile robot sensing, the project evaluates nitrogen use efficiency (NUE) across various cultivars of Lolium perenne (Perennial Ryegrass). To support this goal, two test fields were established in the Norwegian cities of Ås and Hamar. Each field is divided into three nitrogen treatment groups, each further subdivided into three replicates. Within each replicate, 40 ryegrass varieties were randomly planted to minimize location-based bias.

This paper presents a proof-of-concept (PoC) soil monitoring system designed to validate the core functionality of a selected sensor. By implementing this, potential data management challenges and opportunities can be identified early. Given the project's focus on robotics and automation, the system must communicate with an onboard robot computer, leverage the Robot Operating System (ROS) when possible, and support both remote access and local data storage. With those needs, a self-contained PoC solution using soil sensors based on electrical properties is introduced. The goal is to test the technology for further deployment in the experimental field.

A comparative experiment was conducted to evaluate how sensor readings respond to different water treatments applied to soil samples, assessing whether the data aligns with expected outcomes in preparation for field deployment. In addition, an energy consumption analysis was performed to optimize system efficiency and enable future improvements to the control board responsible for sensor management. To complement these tests during the off-season, an additional experiment was carried out in a hydroponic environment to assess system performance in a controlled setting.

2. SOIL MONITORING SYSTEM

As Dattatreya et al. (2024) explains, most agricultural soil sensors can be categorized as electrical or electromagnetic, optical or radiometric, mechanical, acoustic, or electrochemical. Electrical-conductivity-based sensors can be readily found at low prices, however, often there is insufficient or limited documentation about calibration, limitations, and operating principles.

The proposed PoC solution explore three sets of sensors, where each set of sensors is composed of three devices.

Table 1. Sensor measurement parameters.

Sensor	Parameters
Electrical Conductivity and Hydrogen Potential	EC Measuring Range: 0-20000 µS/cm EC Resolution: 1 µS/cm EC Accuracy: 0-10000 µS/cm@±3%FS 10000-20000 µS/cm@±5%FS EC T. Compensation: Range 0-50°C pH Measuring Range: 3-9 pH pH Resolution: 0.1 pH
Nitrogen Phosphorus Potassium	NPK Measuring Range: 0-2999 mg/kg (mg/L) NPK Resolution: 1 mg/kg (mg/L) NPK Accuracy: $\leq$ 5%
Temperature and Soil Humidity	Hm. Measuring Range: 0-100% Hm. Resolution: 0.1% Hm. Accuracy: 0-50%@±2%, 50-100%@±3% T. Measuring Range: -40°C to +80°C T. Resolution: 0.1°C T. Accuracy: ±0.5°C

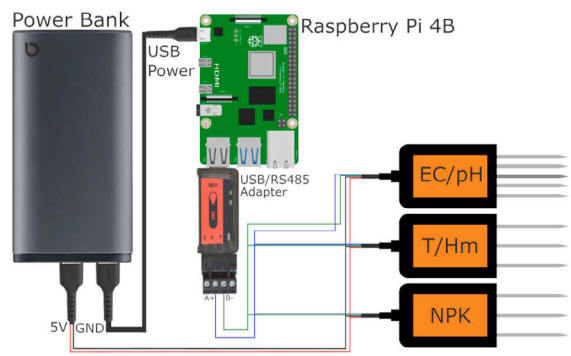

Fig. 1. Proof of Concept diagram for the soil monitoring system - One Battery powering the Raspberry Pi and the sensors while using the USB adapter for sensor communication. All data is outputted wirelessly.

Table 1 shows the relevant characteristics for each sensor in one set. All of them communicate over RS485 through the MODBus-RTU protocol, supplied from $5V$ to $30V$ DC with an alleged power consumption of up to $0.5W$ at $24V$. The sensors are specified at IP68 protection level, being composed of stainless steel probes and a sealed solid epoxy resin body.

This PoC solution is built from off-the-shelf components and systems, primarily focused on ease of use and development simplicity. The main adopted system is a Raspberry Pi 4B, a powerful and well-known single-board computer (SBC) with the capability of running ROS nodes itself. An Ubuntu Server 20.04 image for the Raspberry Pi was chosen due to the ROS 1 support, making it compatible with the available robotic platform. All necessary code was written in Python 3, taking advantage of Pyserial and Pymodbus for sensor communication.

Since RS485 demand voltage levels that are not available on the Raspberry Pi, a module responsible for the communication has to be added. However, to simplify this initial phase, a USB to RS485 adapter was chosen for this task initially, which allows for easier development, enabling tests in a different range of systems. In this evaluation, the system was able to be powered by an off-the-shelf $15000mhA$ power bank. Figure 1 presents a diagram for the proposed PoC solution.

Fig. 2. The 12 soil samples separated for the experiment - Each one composed of 70 grams of dry soil mixture and 100 grams of water.

3. METHODOLOGY

Initially, to evaluate the consistency of the sensor data, 12 small soil samples of potting soil from Norgro (item number 191-100520) were collected and distributed equally by weight, as can be seen in Fig. 2. Stones that could interfere with sensor placement were removed. The soil is a mixture of fine sand and white moss (H2-H7) and is expected to have a pH between 5.5 and 6.5, with nutrient concentrations of 300 mg/L total nitrogen ($NO_3 + NH_4$), 100 mg/L phosphorus, 400 mg/L potassium, and 1500 mg/L calcium.

Each sample consists of 70 grams of dry soil mixed with 100 grams of water placed in impermeable 200-milliliter plastic vessel, thus only losing water through evaporation. Given the consistent volume, weight, and soil origin, the density and composition are expected to be uniform across all samples. The 12 soil samples were divided into 6 pairs (A to F), with each pair receiving a distinct water treatment. Data collection was conducted daily, with each sample receiving extra 20 grams of treatment water. Table 2 lists the different treatments deployed for testing.

Table 2. Water treatments applied to each pair of samples.

Group	Treatment
A	Water directly
B	Water* with default amount of N fertilizer
C	Water* with 15% less N fertilizer
D	Water* with 25% less N fertilizer
E	Commercial Nutrient solution diluted by 2.5:1
F	Commercial Nutrient solution diluted by 5.0:1

The system power consumption was logged with two INA219 DC current monitoring boards, as seen in Fig. 3a. The current and voltage are directly read and combined into a power reading for each monitored rail with $10Hz$ polling frequency. The first board measured power for the current set of sensors, while the second board focused on the Raspberry Pi. The sensor power consumption was also evaluated while idle, without any data communication.

The sensors were grouped by type, creating three sets of three sensors in their out-of-the-box state, as no explicit calibration procedures were provided in the documentation. Each set was placed individually in a sample for approximately 2 minutes, collecting 30 to 40 measurements, starting 3 hours after watering. A general view of the experiment can be seen in Fig. 3b. The sensor probes were cleaned with clean water between measurements. The

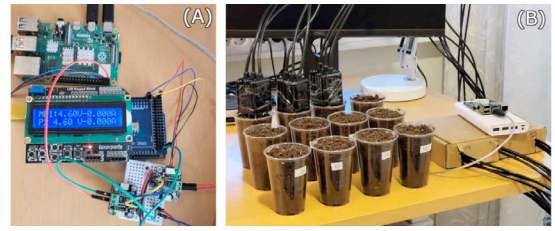

Fig. 3. (A) Energy consumption measurement setup with INA219 board; (B) Sensor and system setup during the data collection phase.

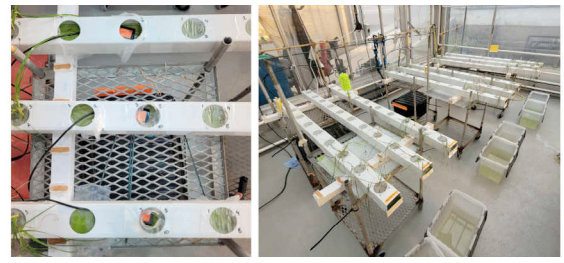

Fig. 4. Hydroponics setup similar to the field experiments - Three nitrogen levels with three rows each.

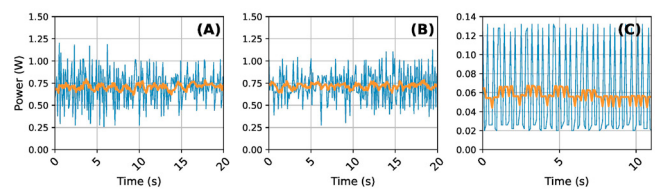

Fig. 5. Sensor power consumption comparison (A) while idle and (B) while receiving a query from the host system each 5 seconds. (C) Power consumption for a single sensor. Orange line denotes the average in a 1 second window.

collected data were locally stored, streamed in real-time to a Google Sheets table as a PoC for a cloud-based visualization solution, and ROS-bag was recorded using a second computer to test data transmission.

The final preliminary test was conducted at the NMBU Biovit greenhouse using the hydroponic setup shown in Fig. 4, allowing for the evaluation of the system performance in a controlled hydroponic environment, with three nitrogen fertilizer levels. During this test, a longer experiment was performed, where the sensors remained in operation for three days while being powered externally.

4. RESULTS

The first notable observation is that those sensors continuously draw current, whether idle or during a read operation. Figures 5a and Fig. 5b show power data for all nine sensors in both scenarios, averaging $0.71\,W$ at $5\,V\,DC$, lower than expected given the available datasheet. This suggests that the sensor energy consumption is independent of the computer's polling interval.

When looking at a single sensor, specifically the NPK sensor, a pattern in energy consumption emerges. In Fig. 5c, short spikes of $0.13\,W$ can be seen with a base consumption of $0.02\,W$, averaging to a power consumption of $0.06\,W$. Note that the logging device might not have a high enough

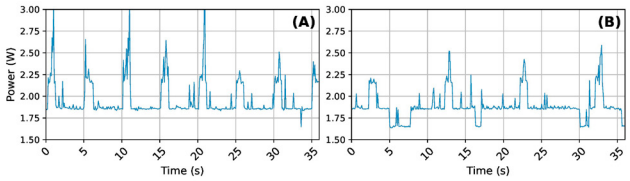

Fig. 6. Comparison between different query and upload rates for the Raspberry Pi: (A) query each 5 seconds; (B) query each 10 seconds.

sampling rate to fully capture the precise behavior of the sensor. For the expected field scenario, one set of three sensors per device, an average consumption of $0.24\,W$ was found, which indicates that all sensors have similar power consumption.

Meanwhile, Fig. 6a and b show the Raspberry Pi power consumption in two different polling scenarios with a display attached: polling every 5 seconds and 10 seconds, respectively. The Raspberry Pi held a base power consumption for most of the collected data with transient spikes. A spike can be seen evenly spaced at the expected polling rate, likely due to the main program loop. The tasks of reading, processing, storing, and uploading the data to a cloud solution cause the Raspberry Pi to hit a momentary peak power consumption, indicating plenty of computational power available.

By averaging the results for each scenario, an energy consumption of $1.95\,W$ is observed when polling the sensor every 5 seconds. Increasing the polling interval to 10 seconds reduces power consumption by 4%, with a further 2% reduction when extending it to 60 seconds. Additional power savings were achieved by unplugging the attached screen, which represents the final intended configuration. In this setup, the Raspberry Pi maintained an average power draw of $1.75\,W$. Including the expected consumption of a single sensor set, the system averages approximately $2.0\,W$ of continuous power draw in its current implementation, with higher short transients.

Fig.,7 shows trends observed across all samples. Some variation was seen among the sensors, however, their relative positions remained generally consistent. Sensors that initially recorded higher or lower values tended to maintain this pattern, with the exception of humidity and pH in sensor one.

By averaging each sensor triad per sample and averaging the twin samples, Fig. 8 can be built. Humidity and temperature values are the more consistent. All pH values start with a larger spread and converge to a similar value. Similarities can be noticed between the NPK values and the conductivity, a clue that could lead us to interpret the NPK values as a value derived from conductivity. A trend for this graph starts to emerge on Day 4 as three distinct groups form: the normal water, the fertilizer, and the samples derived from the hydroponics system.

The values from Fig. 8 can be compared to the ones in Table 3, where the humidity reads near or equal to 100%, as expected for measuring only the treatment water. The discrepancy in temperature values is expected as the different mixtures were stored in different places initially. All registered pH values in Tab. 3 are slightly acidic and

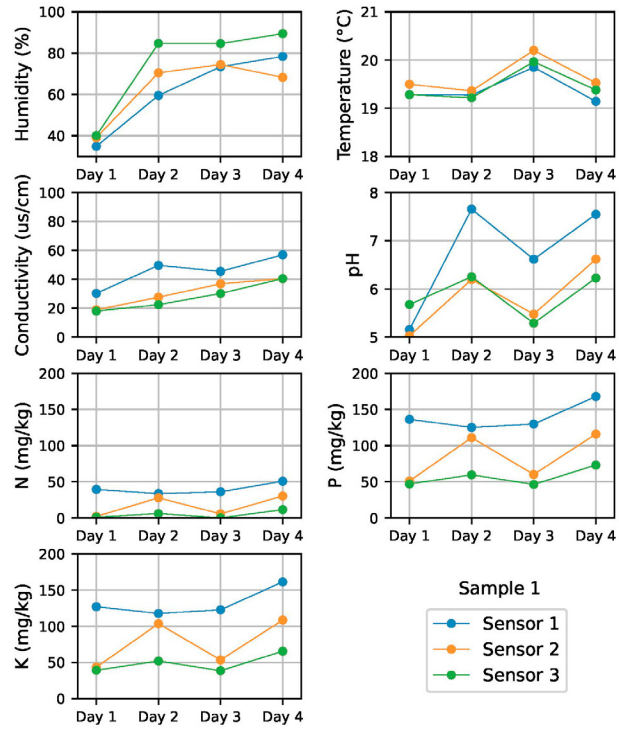

Fig. 7. Measurements across four days for each sensor in a single sample.

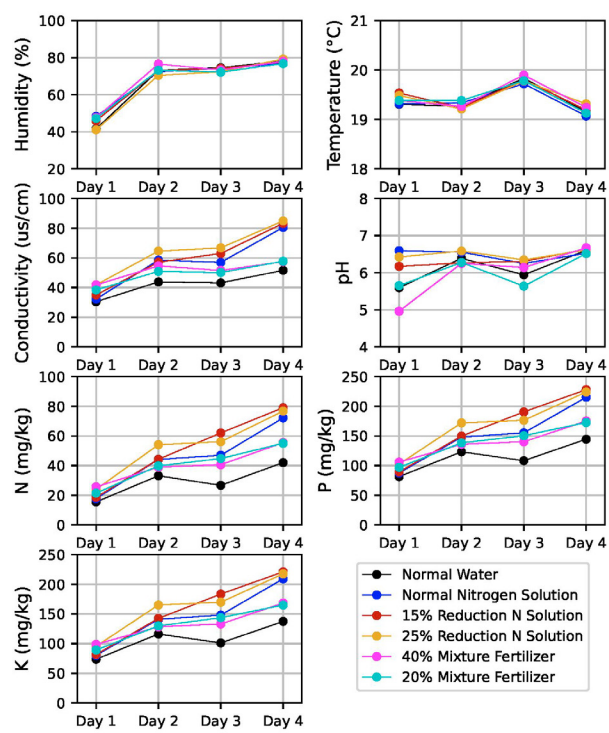

Fig. 8. Measurements across four days for each water treatment with the pairs of samples and sensors averaged.

differ from those observed on Day 4, as shown in Fig. 8. Considering that the expected soil pH ranges between 5.5 and 6.5, these values remain within a reasonable range for the soil. Meanwhile, both N, P, K, and conductivity readings form trends similar to those observed previously.

Table 3. Average of sensor measurements obtained for each water treatment directly.

Treatment Water	Hm %	T °C	EC mS	pH	N mg/L	P mg/L	K mg/L
Water	100	18	6	7	0	0	0
Norm. N	96	19	103	5	102	279	265
-15% N	98	19	103	5	97	269	263
-25% N	95	19	107	5	99	278	273
40% Fert.	99	17	16	4	0	16	8
20% Fert.	100	17	19	4	0	34	26

Finally, Fig. 9 presents the hydroponic system data across the three observed days. Some discrepancies in humidity measurements are noticeable. Given the expected value of 100%, this may be due to improper probe placement, which was an issue during this setup. Temperature trends were consistent across all probes, with probe one (blue) deviating by less than one degree. PH measurements showed a discrepancy, with probe one (blue) differing from the others. This setup is the only scenario where conductivity significantly differs from the estimated N, P, and K values, exhibiting a more stable reading. However, the observed order of values deviates from expectations, once again due to probe one (blue) behaving differently.

The N, P, and K values show similar readings, further reinforcing concerns about the inability to distinguish nutrients. Cyclical nutrient variations were detected on Day 1 with probes two (orange) and three (green), and on Day 2 with probe one (blue), possibly influenced by the pump's activation near the sensor. This suggests a potential capability to detect fluctuations in the water mixture even if inconsistent. By Day 3, as environmental conditions stabilized, sensor readings became more consistent, but probe three (green) showed higher values than probes one (blue) and two (orange).

During the execution of the hydroponic test, the sensors remained functional and unattended for three days without issues, completely submerged. Which indicates that their structure is as robust as claimed. The system board operated successfully, uploading the data as expected and storing the data locally simultaneously.

5. DISCUSSION

The initial hardware implementation performed as expected. The Raspberry Pi enough, but with a runtime limited to approximately 24 hours. To extend operation time when running standalone, a larger battery or a solar power solution must be considered. Preliminary tests with an Arduino board showed sufficient computational capacity when transmitting data via a serial connection to an external computer. The collected data did not require a large storage solution, as local SD card storage proved to be sufficient and presented no limitations. ROSbag recording worked reliably for local logging, while live cloud uploads operated smoothly, requiring minimal bandwidth, storage capacity, and processing power.

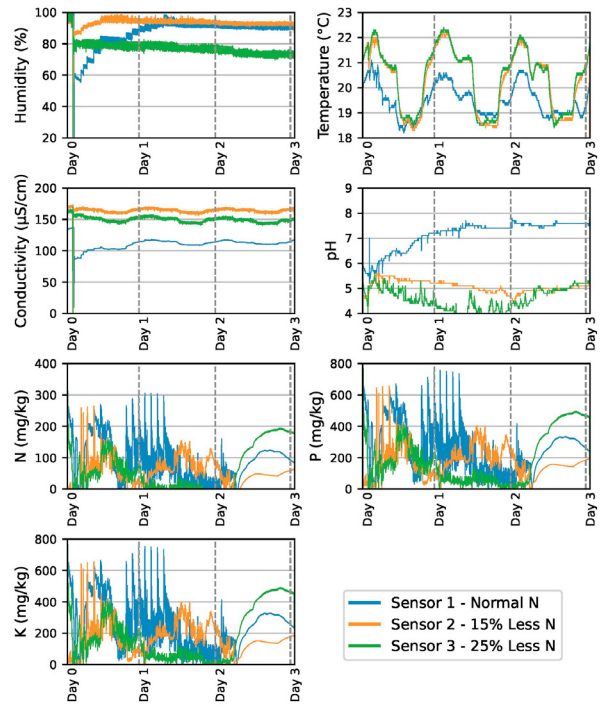

Fig. 9. Measurements for the hydroponic setup.

Sensor power consumption was found to be independent of data queries. Therefore, the best way to reduce power usage is to turn it off when not in use. Given their short initialization time, this approach is feasible. In a microcontroller-based system, while the controller enters a low-power state, the sensor can be completely deactivated, resulting in significant energy savings. As such a wireless-capable microcontroller could add complexity but significantly improve runtime.

The sensor tested did not perform as advertised. While it successfully detected changes in soil humidity and temperature readings were consistent, other sensor outputs require careful interpretation and consideration. Conductivity, N, P, and K readings only demonstrated potential as indirect indicators of overall ion content. The sensor successfully distinguished between sample groups in the soil test, suggesting that a possible correlation with fertilizer levels could be seen in the test field. However, proper calibration is necessary to enhance precision and obtain measurement accuracy.

Some abnormalities observed in the soil experiment may be attributed to the fact that the mixture extracted from the hydroponic system was stored for a month before use. This may have led to changes in water composition due to microbial and algae activity, potentially explaining the similarity in some recorded values.

6. CONCLUSION

While the proposed hardware performed as expected, an important observation is that the sensor draws power continuously, regardless of whether it is actively communicating with the host system. To address this, a simple switching mechanism should be developed to power the sensor only when needed. This would reduce overall energy consumption, at the cost of slightly increased system complexity, and offer finer control over hardware operation.

The sensor's actual power consumption was found to be lower than expected based on its datasheet. Initially, a set of three sensors was expected to consume approximately $1.5\,W$, comparable to the Raspberry Pi's observed consumption of $1.75\,W$. However, measurements showed that the three sensors consumed only around $0.25\,W$ in total, making the Raspberry Pi the primary power consumer. While this is a positive finding, it highlights the need for a smaller and more efficient microcontroller with wireless capabilities in future iterations.

Some uncertainties emerged regarding the sensor data itself. Although prior studies have not questioned consistency across identical sensors, noticeable discrepancies were observed. Additional factors, such as soil density and compaction, may significantly influence sensor readings. When dealing with the hydroponic setup, sensor placement was critical in order to obtain consistent readings. Across treatments, sensor data trends for the NPK ions showed a strong correlation with soil EC. This supports the hypothesis that soil EC alone, or in combination with other sensor readings, could serve as a proxy for overall ion or nutrient content. Future studies should systematically assess these influences to improve measurement reliability, ideally incorporating laboratory soil analysis alongside sensor measurements for calibration and validation.

Further work is needed to refine the system and enhance its reliability and efficiency. The next growing season will provide a valuable testbed for evaluating the sensors in the actual field environment. By integrating this remote sensing technology into mobile robots and drone-based data collection platforms, the DLT-Farming project will be able to advance towards its vision of developing a data-driven agricultural monitoring system to provide informed decisions for farmers.

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